SOC Alert to Response Simulation Report

VSFTPD Exploitation using Metasploit

1. Introduction

This project demonstrates a complete Security Operations Center (SOC) workflow, including attack simulation, detection, analysis, and response. The attack was performed using Kali Linux as the attacker machine and Metasploitable as the target system.

The objective is to understand how real-world cyberattacks are identified, analyzed, and mitigated using standard SOC practices.

Tools Used:

- Kali Linux (Attack simulation)
- Metasploit Framework (Exploitation)
- Metasploitable (Target system)
- Netstat (Evidence collection)

Note: SIEM tools like Wazuh and TheHive are typically used for alert monitoring and incident management in real SOC environments.

2. Network Configuration

The first step involves identifying the IP addresses of both attacker and target machines to ensure proper communication within the virtual environment.

```
┌──(root㉿kali)-[~]
└─# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:16:ff:97 brd ff:ff:ff:ff:ff:ff
    inet 192.168.19.129/24 brd 192.168.19.255 scope global dynamic eth0
       valid_lft 1777sec preferred_lft 1777sec
    inet6 fe80::20c:29ff:fe16:ff97/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

Figure-1: IP address configuration of attacker machine (Kali Linux), showing active network interface and assigned IP used to launch the attack.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:37:03:42
          inet addr:192.168.19.128  Bcast:192.168.19.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe37:342/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:112 errors:0 dropped:0 overruns:0 frame:0
          TX packets:88 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:8630 (8.4 KB)  TX bytes:10653 (10.4 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:122 errors:0 dropped:0 overruns:0 frame:0
          TX packets:122 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:33885 (33.0 KB)  TX bytes:33885 (33.0 KB)

msfadmin@metasploitable:~$
```

Figure-2: IP address configuration of target machine (Metasploitable), identifying the vulnerable system used in the exploitation process.

3. Attack Execution

The attack was performed using the Metasploit Framework by exploiting a known vulnerability in VSFTPD version 2.3.4 (vsftpd_234_backdoor). This vulnerability contains a backdoor that allows attackers to execute commands remotely.

The Metasploit module was configured with:

- RHOST: Target machine (Metasploitable)
- LHOST: Attacker machine (Kali Linux)

Upon execution, a shell session was successfully opened, resulting in root-level access to the target system. This confirms successful remote exploitation.

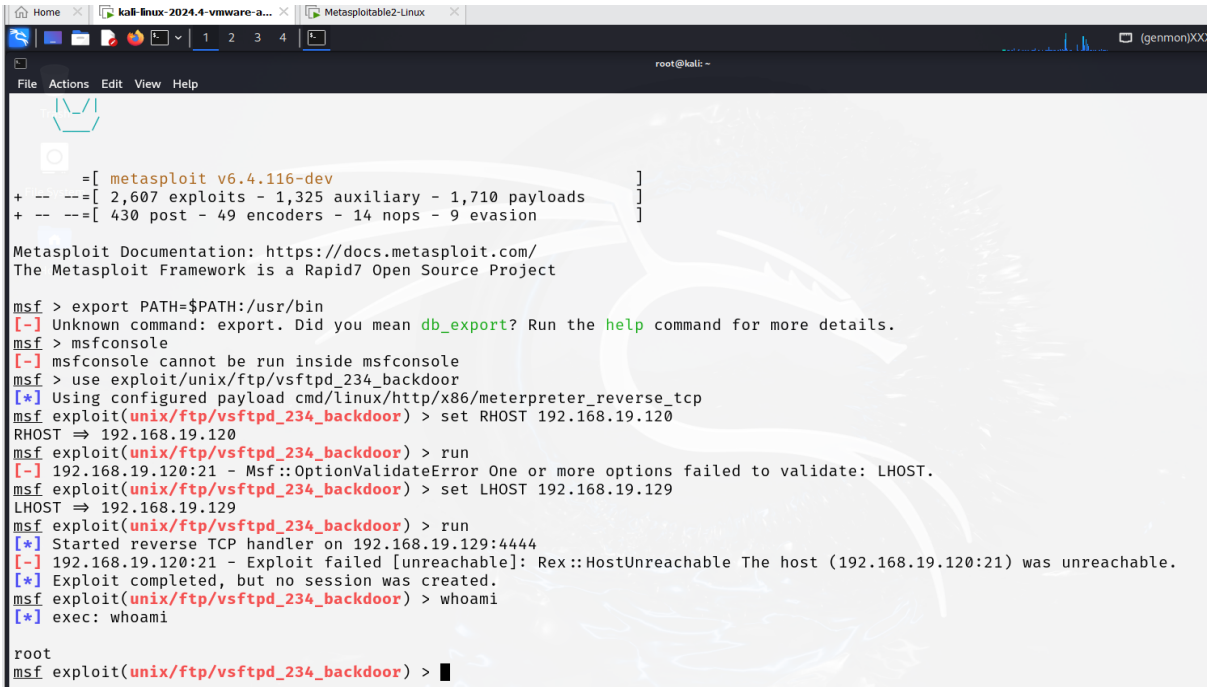

Figure-3: Configuration and execution of the VSFTPD backdoor exploit using Metasploit Framework, demonstrating successful remote access and privilege escalation on the target system.

4. Alert Details

Alert ID	Description	Source IP	Target IP	Priority	Status
001	VSFTPD Exploit	192.168.19.129	192.168.19.120	Critical	open

The alert is classified as Critical due to successful remote exploitation resulting in unauthorized root access. This indicates a severe compromise with high impact on system security.

5. Incident Classification

The incident is classified as:

- Type: Remote Exploitation
- Category: Unauthorized Access
- MITRE Technique: *T1190* (Exploit Public-Facing Application)

Metadata:

- **Source IP:** *192.168.19.129*
- **Target IP:** *192.168.19.120*
- **Service:** *FTP (VSFTPD)*
- **Indicator:** *Backdoor exploit*

6. MITRE ATT&CK Mapping

Using MITRE ATT&CK, the attack is mapped to:

- **Technique ID:** *T1190*
- **Technique Name:** Exploit Public-Facing Application

Explanation: The attacker exploited a vulnerable FTP service (VSFTPD) to gain unauthorized access to the system. This aligns with the MITRE ATT&CK technique T1190, where public-facing applications are targeted.

6.1 Alert Triage

The alert was analyzed based on severity, indicators, and system impact.

- Alert ID: 001
- Description: VSFTPD Exploit Detected
- Source IP: 192.168.19.129 (Attacker – Kali Linux)
- Target IP: 192.168.19.120 (Metasploitable)
- Priority: Critical
- Status: Confirmed Incident

- ✓ The alert indicators were conceptually validated using threat intelligence sources such as VirusTotal and AlienVault OTX to confirm malicious activity.
- ✓ The alert was validated as a true positive due to successful exploitation of the vulnerable VSFTPD service, resulting in unauthorized root-level access.
- ✓ Based on the severity and impact, the incident was escalated to higher-level SOC analysts (Tier 2) for further investigation, containment, and remediation.

7. Alert Priority Justification

The alert is classified as Critical because the attacker successfully gained root-level access to the system. This indicates a complete system compromise affecting confidentiality, integrity, and availability.

The alert priority is determined using standard SOC classification methods. Based on CVSS scoring, the VSFTPD vulnerability has a high severity score (approx. 9+), indicating critical risk.

Factors Considered:

- Asset Criticality: Target system is vulnerable server
- Exploit Availability: Public exploit exists in Metasploit
- Impact: Root-level access leads to full compromise

Hence, the alert is classified as Critical.

8. Incident Response Actions

The incident response followed standard NIST lifecycle:

- Identification: Detection of VSFTPD exploit activity
- Containment: Isolated the compromised system
- Eradication: Removed vulnerable service
- Recovery: Planned system restoration
- Lessons Learned: Importance of patching and monitoring

9. Evidence Collection

Evidence was collected using system-level commands to capture active network connections during the attack.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > netstat -antp > evidence.txt
[*] exec: netstat -antp > evidence.txt

msf exploit(unix/ftp/vsftpd_234_backdoor) > sha256sum evidence.txt
[*] exec: sha256sum evidence.txt

f64ef28b1f19b7b03549a57a2e57327fc124c5bcfbfffadd9c2cd69f17b94fc0  evidence.txt
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

Figure-4: Evidence collection and integrity verification using netstat and SHA256 hashing.

Evidence Details:

- Evidence: Network connections log
- Method: netstat command
- Hash Value: *94557aeefaada3c97730020780b3c8c08d40e7ed8fd83c001e8fdd8491c6d2da*

This ensures the integrity of the collected evidence using SHA256 hashing.

10. Incident Timeline

- 11:00 – Attack initiated from Kali Linux
- 11:02 – VSFTPD exploit executed
- 11:03 – Root access gained on target
- 11:05 – Evidence collected
- 11:07 – System isolated

11. Capstone Project: Full Alert-to-Response Cycle

This project demonstrates a complete Security Operations Center (SOC) workflow from attack simulation to incident response.

- Attack Simulation: A VSFTPD backdoor vulnerability was exploited using Metasploit Framework against the Metasploitable target system.
- Detection: Suspicious activity was identified through monitoring of service exploitation and unauthorized access attempts (simulated SOC detection).
- Triage: The alert was analyzed based on severity and indicators, and classified as Critical due to successful exploitation and root-level access.
- Response: The affected system was logically isolated, and the attacker machine was identified and blocked to prevent further access.
- Evidence Collection: Network connection logs were collected using netstat and integrity was verified using SHA256 hashing.

This end-to-end process validates the SOC lifecycle, including detection, analysis, response, and documentation of a security incident.

12. Impact Analysis

The attacker gained root-level access to the target system, allowing full control over system operations. This could lead to data breaches, system manipulation, and further network compromise. The impact is considered severe.

13. Recommendations

- Update and patch vulnerable services
- Disable unnecessary services like FTP
- Implement firewall rules
- Deploy intrusion detection systems
- Perform regular security assessments
- Implement SIEM solutions like Wazuh for real-time monitoring

14. Conclusion

This project successfully demonstrated a complete SOC workflow, including attack simulation, alert detection, triage, and incident response. The VSFTPD vulnerability was exploited using the Metasploit Framework, and the attack was analyzed using MITRE ATT&CK (T1190) for proper classification.

The alert was prioritized as Critical based on its impact and successful root-level compromise. Evidence was collected and verified using hashing techniques to maintain integrity.

This project highlights the importance of continuous monitoring, timely patching, and structured incident response procedures in minimizing security risks and improving organizational security posture.